

Lab 4. Managing User Accounts.

IT Infrastructure Management (W04IST–SI0826G)

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1 Lab Objectives

1. Learn how to manage user accounts and groups.
2. User profiles.
3. Configure user groups.

2 Introduction

Every computer user must have an avatar (an avatar, a digital identity) in the operating system. Based on this avatar, the operating system decides whether they can access resources. A user account is precisely such an avatar. It is a collection of settings describing the user's characteristics: first name, last name, address, phone number, groups to which they belong, etc. Additionally, the account also has a method for verifying whether the person using the computer has access to the account. This can be a login (login name) combined with a password, fingerprint, or PIN, or a chip card with a PIN.

The operating system identifies an account by its Security Identifier (SID). In the account settings, you can freely change the username, login, and descriptions, and it will still be the same account. At the same time, if we delete an account and recreate it, even though the same names, logins, and passwords are set, it will be a completely different account, because a new SID is created when the account is created.

Many settings are associated with an account. Some of them (names, addresses, group memberships) are stored directly in Active Directory. Other data, such as the user's files, are stored in a profile. A profile is a directory stored by Windows in the **C:\Users** directory. Upon first login, if the user does not have a profile, it is created based on a template stored in the **C:\Users\Default** folder. If the user logs in to different computers, they will have a separate profile by default on each one. It is also possible to configure a roaming profile, which is stored on the server. This profile is downloaded to the local computer each time the user logs in, and saved back to the server upon logout. Profile contents are always copied, not synchronized, so if a user logs out of multiple computers, the profile contents match the profile of the computer from which the user logged out last.

3 Exercises

Before the exercise, refer to the book “Exam ref 70-742 Identity with Windows Server 2016” on [Safari Bookshelf](#) and review the sections “Skill 1.2: Create and manage Active Directory users and computers” and “Skill 1.3: Create and manage Active Directory groups and organizational units” in “Chapter 2. Manage and maintain AD DS”.

3.1 User accounts.

1. Create user accounts in various organizational units using the following methods:

- Active Directory Administrative Center (ADAC) console,
- Active Directory Users and Computers (ADUC) console,
- Command line – `dsadd user` (hint: `dsadd user /?`),
- PowerShell – `New-ADUser` (hint: `get-help New-ADUser -examples` or `-online`).

When creating accounts, use the following login convention:

- UPN (User Principal Name) - `firstname.lastname@contoso.com` (note that the Active Directory domain has a DNS name: `contoso.internal`, and the UPN, which is the email address, has a DNS name: `contoso.com`)
- Pre-Windows 2000 (SAMid) - `contoso\inlastname` (first name + last name)

Set both the UPN and SAMid in the administrative consoles, as well as in CMD and PowerShell (search for options in the prompts and help files).

When creating an account, remember to set a password. Note that accounts created without a password will be automatically disabled.

Accounts created using PowerShell are automatically disabled unless they also have the `-Enabled:$True` option. The password must be specified as `SecureString`, but you can convert it from plain text using the command: `(ConvertTo-SecureString -AsPlainText MyPassword1`

1. Test different username combinations for the selected account and see which ones work. Connect to machine CL1 using remote desktop, specifying a new user account each time:

- `firstname.lastname@contoso.com`
- `lastname@contoso.com`
- `contoso\firstname.lastname`
- `contoso\fname`

Which attempts were successful?

1. Modify the account settings (e.g., add a phone number, office address, home address) using the following commands:

- Command Line (CMD, command line) – `dsmod`
- PowerShell – `Set-ADUser`

1. As part of the modification, test disabling and enabling the account. Check what message you receive when trying to log in to the disabled account.

2. Log in to machine CL1 with one of your accounts, entering the incorrect password four times. The account should be locked on the last attempt. See what message appears on the login screen. Unlock the account using the ADAC and ADUC consoles.
3. Test which fields from the user account are used for display (you can create a new account by assigning different values to the Name and Display Name fields):
 - username at login,
 - in the Start menu next to the user's avatar,
 - in the user list in the ADUC and ADAC consoles.
1. Disable several accounts in different organizational units. Search for disabled accounts using the following tools:
 - ADUC and ADAC consoles,
 - CMD console: **dsget**,
 - PowerShell console: **Get-ADUser**.

How will you limit the search scope to the selected organizational unit in each of these cases?

4 Roaming Profiles

Note: On machine DC1, the **C:\profiles** folder (visible on the network as **\\dc1.contoso.internal\profiles**) is already configured and shared for storing profiles.

1. Select one of the previously created accounts. Add them to the **Domain Admins** group (so the account can log in to DC1). In the **Profile** field, set the profile path:
 - In the ADAC console, enter the path with the username, e.g.:

\\dc1.contoso.internal\profiles\<username>

- In the ADUC console, try the path (enter **%USERNAME%** as is):

\\dc1.contoso.internal\profiles\%USERNAME% .

Note what happens when you enter this path and press the **Apply** button.

1. In the ADUC console, copy the account where you set the path with the **%USERNAME%** variable. Look at the profile path in the copied account. What changed?
2. Log in to DC1 using the account with the profile. Change the desktop wallpaper. Place the files on the desktop. Create files in the **Documents** folder. **Log out** (Sign out) from DC1.
3. Log in to machine CL1 using the same account. Did the settings transfer from DC1? Change the desktop color and **log out**.
4. Log back into DC1 and check if your desktop color is the same as on CL1.

5 Groups

1. Create user groups using the following tools:

- ADAC and ADUC consoles,
- CMD console,
- PowerShell console.

Create a different group type each time (domain local, universal, global). What group type and scope are selected by default?

1. Using the ADUC console, check whether:

- you can nest groups within each other (e.g., a global group within a global group)?
- can you add a global group to a universal group? A domain local group to a universal group? Test which combinations are acceptable.

1. Add a user to a group using CMD (dsmod) and PowerShell (Add-ADGroupMember).

2. Add one group to another using CMD (dsmod) and PowerShell (Add-ADGroupMember).